HOW IT WORKS

The loop, end to end.

Drone camera
streams over its own WiFi

Laptop: find every person
YOLO11n + ByteTrack

Which one is you?
ArcFace face + body re-ID

Turn it into a setpoint
Kalman + PD, no integral

30-byte packet, CRC-16
30 Hz, sent 3x

ESP32-S3 keeps it flying
1 kHz attitude loop

the aircraft moves, so the camera moves, so the picture changes

THE HARD PART

Two signals, different jobs.

A face confirms it is you
ArcFace 512-d · strict · threshold 0.36

But only 26% of frames have one
you turn, you blur, you are too far away

So the body carries the other 74%
768-d re-identification · sustains, never acquires

And a WRONG face vetoes the body
below 0.15 it counts as evidence AGAINST — without that, two people in similar clothes swap identity forever

Verified on two real people: 98% locked, 0 transfers

ON THE AIRCRAFT

Sense, decide, act. A thousand times a second.

IMU, 1 kHz
one burst so every axis shares an instant

Madgwick filter
gyro + accel into an attitude estimate

Angle loop, then rate loop
P outside, PID inside, D on measurement

Mixer
shifts all four motors, never clips one

Staggered PWM
4 motors on 4 different edges, not one 11 A spike

No command for 400 ms: level off. For 3 s: land.

It never holds the last thing you told it.